

Lemma. Let I be a proper ideal of an integral domain R .

If a non-constant monic polynomial $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ satisfies

$$\overline{P(x)} = x^n + (\overline{a_{n-1}} + I)x^{n-1} + \dots + (\overline{a_1} + I)x + (\overline{a_0} + I) \in (R/I)[x]$$

cannot be factored two smaller degree polynomials ($\Leftrightarrow$ irreducible)

then $P(x)$ is 'irreducible' in $R[x]$.

Proof. Suppose that $\overline{P(x)}$ cannot be factored and $P(x)$ is reducible.

That is, $P(x) = A(x) \cdot B(x)$ where $A(x), B(x)$ are non-constant monic polynomials.

Then, $\overline{A(x)}$ and $\overline{B(x)}$ is the factor of $\overline{P}$, it is contradiction.

Eisenstein Criterion.

Let P be a prime ideal in the integral domain R .

Suppose that a non-constant monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in R[x]$$

satisfies $a_{n-1}, \dots, a_1, a_0 \in P$ but $a_0 \notin P^2$, then f is 'irreducible'.

Proof. Suppose that $f(x)$ is reducible, factored as $f(x) = A(x)B(x)$,

where $A(x)$ and $B(x)$ are monic, degree smaller than $f(x)$. Denote

$$A(x) = x^r + \alpha_{r-1}x^{r-1} + \dots + \alpha_1x + \alpha_0 \quad \text{and}$$

$$B(x) = x^s + \beta_{s-1}x^{s-1} + \dots + \beta_1x + \beta_0. \quad \text{Then, by the assumption,}$$

$$\overline{f(x)} = x^n + (\overline{a_{n-1}} + P)x^{n-1} + \dots + (\overline{a_0} + P) = x^n = \overline{A(x)} \overline{B(x)}.$$

Since P is prime, R/P is again the integral domain,

so either $\overline{\alpha_0}$ or $\overline{\beta_0}$ is zero. WLOG, let $\overline{\alpha_0} = 0$.

Then, $\overline{\alpha_1} \cdot \overline{\beta_0}$ must be zero, as the unique degree 1 coefficient of $\overline{f}$,

so $\overline{\alpha_1}$ or $\overline{\beta_0}$ is zero. If $\overline{\beta_0}$, then $\alpha_0, \beta_0 \in P \Rightarrow \alpha_0\beta_0 \in P^2$, contradiction

If $\overline{\alpha_1} = 0$, then similarly, $\overline{\beta_0} = 0$ or $\overline{\alpha_2} = 0$. Inductively, if $\overline{\alpha_{r-1}} = \dots = \overline{\alpha_1} = \overline{\alpha_0} = 0$,

then $1 \cdot \overline{\beta_0} = 0$, so, $\overline{\beta_0} = 0$, or $\beta_0 \in P$. Now, $\alpha_0\beta_0 \in P^2$, thus proved.